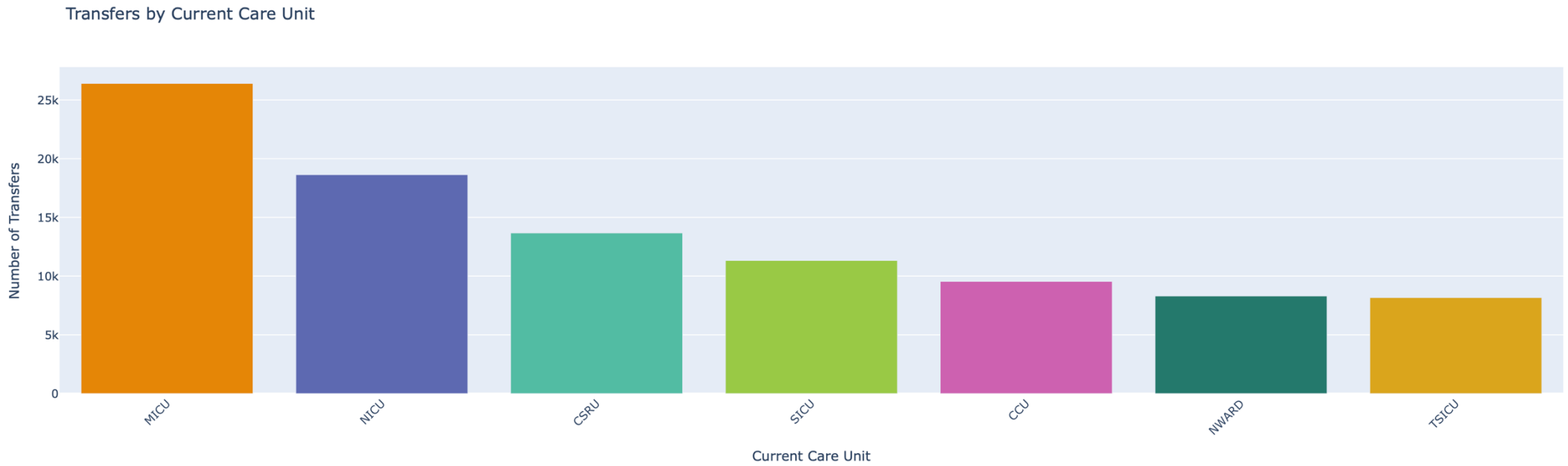
AI in Healthcare Assignment 1

Luke Profio

Plot 1 – Transfers by Current Care Unit



Plot 1 – Transfers by Current Care Unit

- For this chart, we use the TRANSFERS.csv file.
- First, we count how many transfers exist for each care unit.
- Then, we plot a vertical bar chart showing the number of transfers for each care unit.
- This chart shows which care units have the most patient transfers.

Plot 1 – Transfers by Current Care Unit

```
transfer_counts = transfers['CURR_CAREUNIT'].value_counts().reset_index()

transfer_counts.columns = ['care_unit', 'count']

fig1 = px.bar(

transfer_counts,

x='care_unit',

y='count',

color='care_unit',

title='Transfers by Current Care Unit',

color_discrete_sequence=px.colors.qualitative.Vivid

)

fig1.update_layout(

xaxis_title='Current Care Unit',

yaxis_title='Number of Transfers',

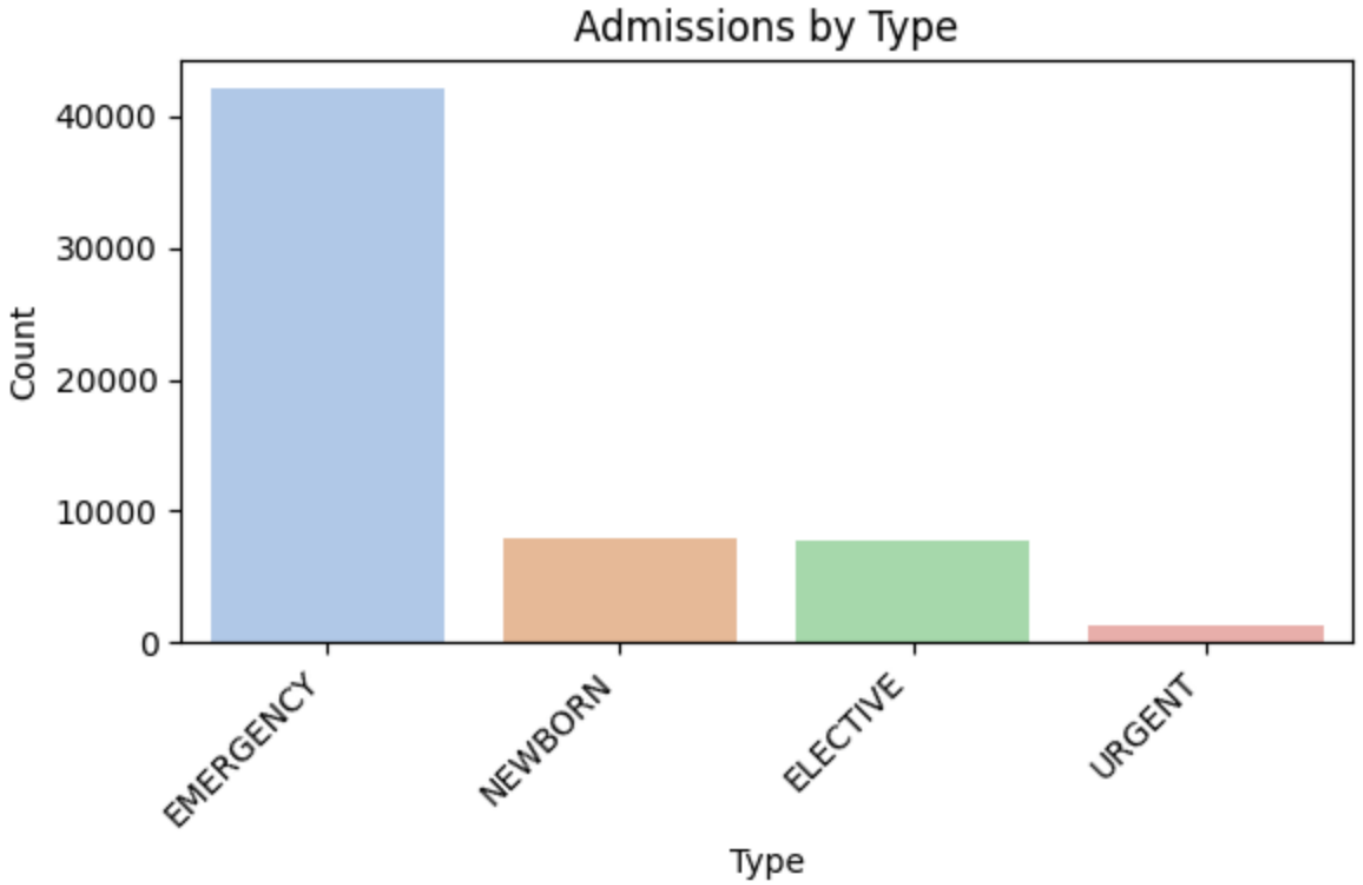
showlegend=False,

xaxis_tickangle=-45

)
```

- `fig1.show()`

Plot 2 – Admissions by Type



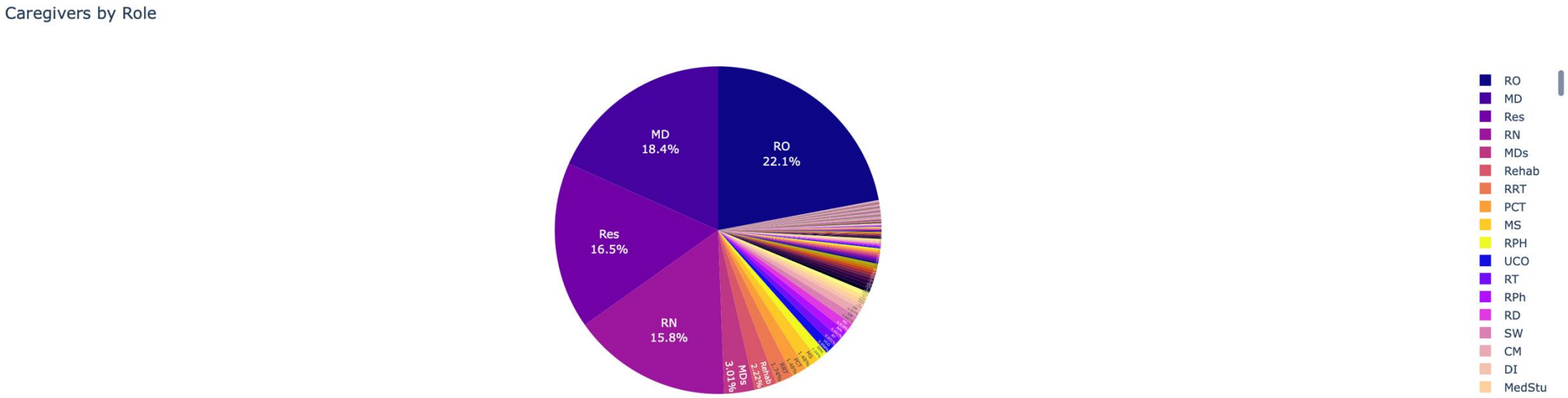
Plot 2 – Admissions by Type

- For this chart, we use the ADMISSIONS.csv file.
- First, we count how many admissions fall under each type, sorting types by descending frequency so the most common ones are listed first.
- Then, we plot a bar chart of admissions for each type.
- This chart shows patients' admission pathway through the hospital.

Plot 2 – Admissions by Type

```
plt.figure(figsize=(6,4))
sns.countplot(
x='ADMISSION_TYPE',
data=admissions,
order=admissions['ADMISSION_TYPE'].value_counts().index,
palette='pastel'
)
plt.title('Admissions by Type')
plt.xlabel('Type')
plt.ylabel('Count')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

Plot 3 – Caregivers by Role



Plot 3 – Caregivers by Role

- For this chart, we use the CAREGIVERS.csv file.
- First, we compute the percentage of records for each role (e.g. MD for physicians).
- Next, we plot a pie chart which displays the percentage of each role.
- This chart shows hospital staffing from a high-level.

Plot 3 – Caregivers by Role

```
care_counts = caregivers['LABEL'].value_counts().reset_index()

care_counts.columns = ['role', 'count']

fig3 = px.pie(

care_counts,

names='role',

values='count',

title='Caregivers by Role',

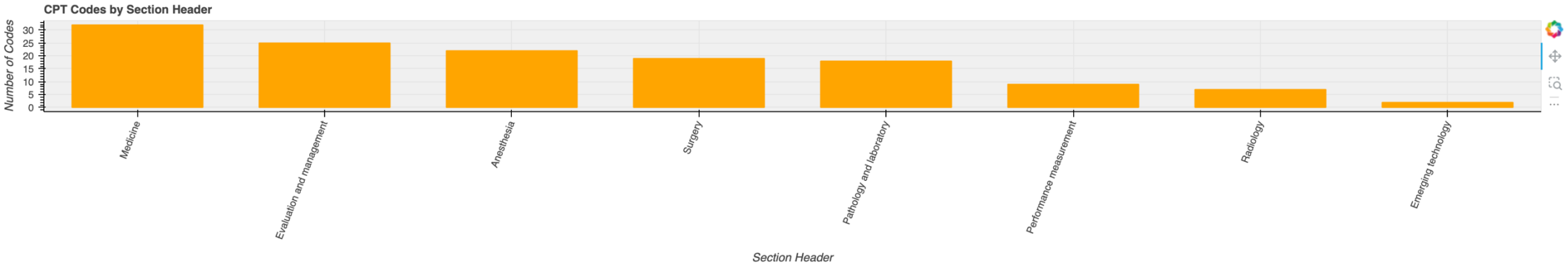
color_discrete_sequence=px.colors.sequential.Plasma

)

fig3.update_traces(textposition='inside', textinfo='percent+label')

fig3.show()
```

Plot 4 – CPT Codes by Section Header



Plot 4 – CPT Codes by Section Header

- For this chart, we use the D_CPT.csv file.
- First, we count how many CPT codes fall under each section header (e.g. evaluation), preserving the natural order of the headers.
- Next, we create a bar chart displaying the counts of the CPT codes for each section header.
- This chart shows section headers with the most CPT codes.

Plot 4 – CPT Codes by Section Header

```
section_counts = d_cpt['SECTIONHEADER'].value_counts().reset_index()

section_counts.columns = ['section', 'count']

source_cpt = ColumnDataSource(section_counts)

p2 = figure(

x_range=section_counts['section'].tolist(),

title='CPT Codes by Section Header',

height=300,

sizing_mode='stretch_width',

background_fill_color='#f0f0f0'

)

p2.vbar(x='section', top='count', width=0.7, source=source_cpt, color='orange')

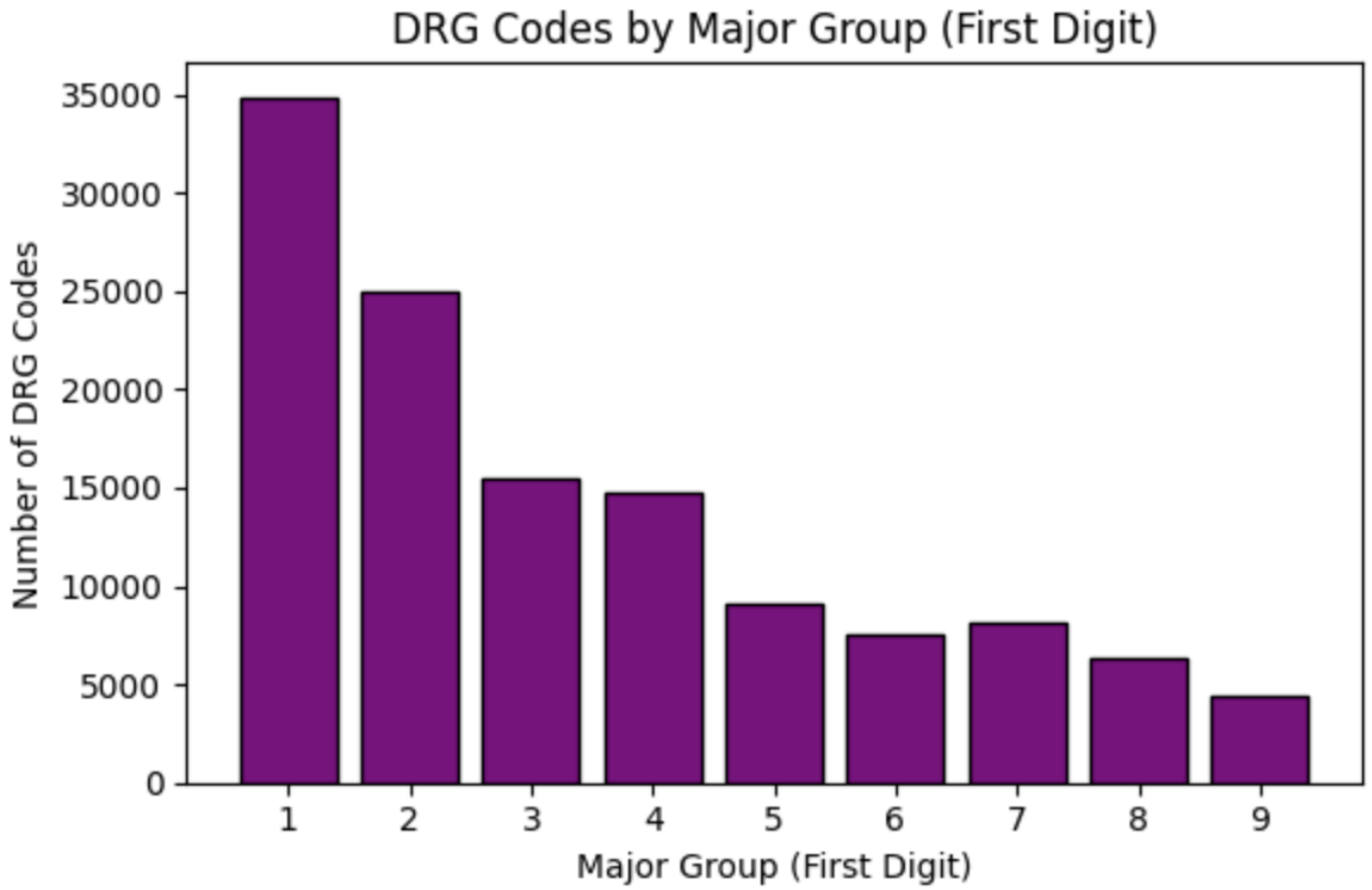
p2.xaxis.major_label_orientation = 1.2

p2.xaxis.axis_label = 'Section Header'

p2.yaxis.axis_label = 'Number of Codes'

show(p2)
```

Plot 5 – DRG Codes by Major Group



Plot 5 – DRG Codes by Major Group

- For this chart, we use the DRGCODES.csv file.
- First, we count how many of the DRG codes belong to each major group.
- Then, we plot a bar chart with one bar representing each major code’s count of DRG codes.
- This chart reveals which major groups have the most DRG codes.

Plot 5 – DRG Codes by Major Group

```
drg['major_group'] = drg['DRG_CODE'].astype(str).str[0]

drg_counts = drg['major_group'].value_counts().sort_index()

plt.figure(figsize=(6,4))

plt.bar(drg_counts.index, drg_counts.values, color='purple', edgecolor='black')

plt.title('DRG Codes by Major Group (First Digit)')

plt.xlabel('Major Group (First Digit)')

plt.ylabel('Number of DRG Codes')

plt.tight_layout()

plt.show()
```